

Top genes by P-value Permulated

Gene	Rho	N	P	p.adj	permPValue	qValueNoPerm	qValuePerm
PTCD1	0.2793375	164	2.920e-04	9.991e-01	2.171e-03	9.999e-01	1.000e+00
LINC00514	0.2827197	123	1.533e-03	9.991e-01	2.629e-03	9.999e-01	1.000e+00
ATP5MF-PTCD1	0.2865243	159	2.258e-04	9.991e-01	2.857e-03	9.999e-01	1.000e+00
MCC	0.3098903	140	1.955e-04	9.991e-01	2.971e-03	9.999e-01	1.000e+00
MFAP5	-0.2836989	124	1.408e-03	9.991e-01	3.657e-03	9.999e-01	1.000e+00
CADPS	0.3775398	86	3.390e-04	9.991e-01	3.885e-03	9.999e-01	1.000e+00
KRTAP10-1	0.1987248	135	2.086e-02	9.991e-01	4.228e-03	9.999e-01	1.000e+00
TCAM1	-0.2925997	144	3.725e-04	9.991e-01	5.029e-03	9.999e-01	1.000e+00
GCM1	-0.2582119	147	1.590e-03	9.991e-01	5.599e-03	9.999e-01	1.000e+00
CYP2W1	-0.2339658	140	5.399e-03	9.991e-01	6.171e-03	9.999e-01	1.000e+00
SPRR2B	0.3210103	79	3.921e-03	9.991e-01	7.429e-03	9.999e-01	1.000e+00
ABL2	0.2656947	143	1.340e-03	9.991e-01	7.656e-03	9.999e-01	1.000e+00
CLIP1	0.2715619	165	4.182e-04	9.991e-01	8.685e-03	9.999e-01	1.000e+00
ZC3H13	-0.2394264	158	2.447e-03	9.991e-01	8.799e-03	9.999e-01	1.000e+00
SCFD2	0.2397081	158	2.417e-03	9.991e-01	8.913e-03	9.999e-01	1.000e+00
KCNH3	0.2397794	144	3.796e-03	9.991e-01	9.028e-03	9.999e-01	1.000e+00
DPM1	0.2662007	129	2.294e-03	9.991e-01	9.370e-03	9.999e-01	1.000e+00
KIF15	-0.1857470	159	1.907e-02	9.991e-01	1.006e-02	9.999e-01	1.000e+00
OTUD7B	0.2699077	133	1.679e-03	9.991e-01	1.017e-02	9.999e-01	1.000e+00
RINL	0.2066348	155	5.890e-03	9.991e-01	1.029e-02	9.999e-01	1.000e+00
KRTAP4-12	0.2708312	94	8.285e-03	9.991e-01	1.051e-02	9.999e-01	1.000e+00
SLC22A8	-0.1881628	148	2.201e-02	9.991e-01	1.063e-02	9.999e-01	1.000e+00
TEX10	-0.2230786	136	9.041e-03	9.991e-01	1.086e-02	9.999e-01	1.000e+00
CIT	0.2523954	154	1.589e-03	9.991e-01	1.097e-02	9.999e-01	1.000e+00

Top genes by P-value non-permulated

Gene	Rho	N	P	p.adj	permPValue	qValueNoPerm	qValuePerm
A1BG	-0.111644190	123	2.189e-01	1.986e-01	9.999e-01	9.999e-01	1.000e+00
A1CF	-0.024569127	128	7.831e-01	9.991e-01	7.988e-01	9.999e-01	1.000e+00
A2M	-0.160699562	169	3.687e-02	9.991e-01	6.502e-02	9.999e-01	1.000e+00
A2ML1	0.052890311	134	5.439e-01	9.991e-01	4.221e-01	9.999e-01	1.000e+00
A3GALT2	0.013537910	132	8.776e-01	9.991e-01	8.713e-01	9.999e-01	1.000e+00
AGALT	-0.065096551	154	4.225e-01	9.991e-01	4.917e-01	9.999e-01	1.000e+00
AGNT	0.054745291	151	5.043e-01	9.991e-01	4.707e-01	9.999e-01	1.000e+00
AAAS	-0.007388264	132	9.330e-01	9.991e-01	9.663e-01	9.999e-01	1.000e+00
AACS	0.019104271	156	8.129e-01	9.991e-01	7.752e-01	9.999e-01	1.000e+00
AADAC	-0.065142431	107	5.050e-01	9.991e-01	5.055e-01	9.999e-01	1.000e+00
AADACL2	0.019040428	120	8.365e-01	9.991e-01	7.951e-01	9.999e-01	1.000e+00
AADACL3	0.024884968	158	7.563e-01	9.991e-01	7.273e-01	9.999e-01	1.000e+00
AADACL4	-0.023590944	141	7.813e-01	9.991e-01	8.335e-01	9.999e-01	1.000e+00
AADAT	-0.058886959	152	4.711e-01	9.991e-01	5.047e-01	9.999e-01	1.000e+00
AGAB	-0.010550651	136	9.030e-01	9.991e-01	9.834e-01	9.999e-01	1.000e+00
AAK1	0.140113651	145	9.278e-02	9.991e-01	2.111e-01	9.999e-01	1.000e+00
AAMDC	0.138093817	88	1.995e-01	9.991e-01	3.761e-01	9.999e-01	1.000e+00
AAMP	0.242451740	95	1.792e-02	9.991e-01	9.702e-02	9.999e-01	1.000e+00
AANAT	-0.106992984	138	2.117e-01	9.991e-01	3.568e-01	9.999e-01	1.000e+00
AAR2	0.188222061	133	3.004e-02	9.991e-01	4.217e-02	9.999e-01	1.000e+00
AARD	-0.111535350	105	2.573e-01	9.991e-01	5.427e-01	9.999e-01	1.000e+00
AARS1	-0.064849571	145	4.130e-01	9.991e-01	5.875e-01	9.999e-01	1.000e+00
AARS2	-0.031408831	165	6.888e-01	9.991e-01	6.514e-01	9.999e-01	1.000e+00
AARSD1	0.007615327	145	9.276e-01	9.991e-01	9.879e-01	9.999e-01	1.000e+00
AASDH	0.090375908	165	2.483e-01	9.991e-01	3.356e-01	9.999e-01	1.000e+00

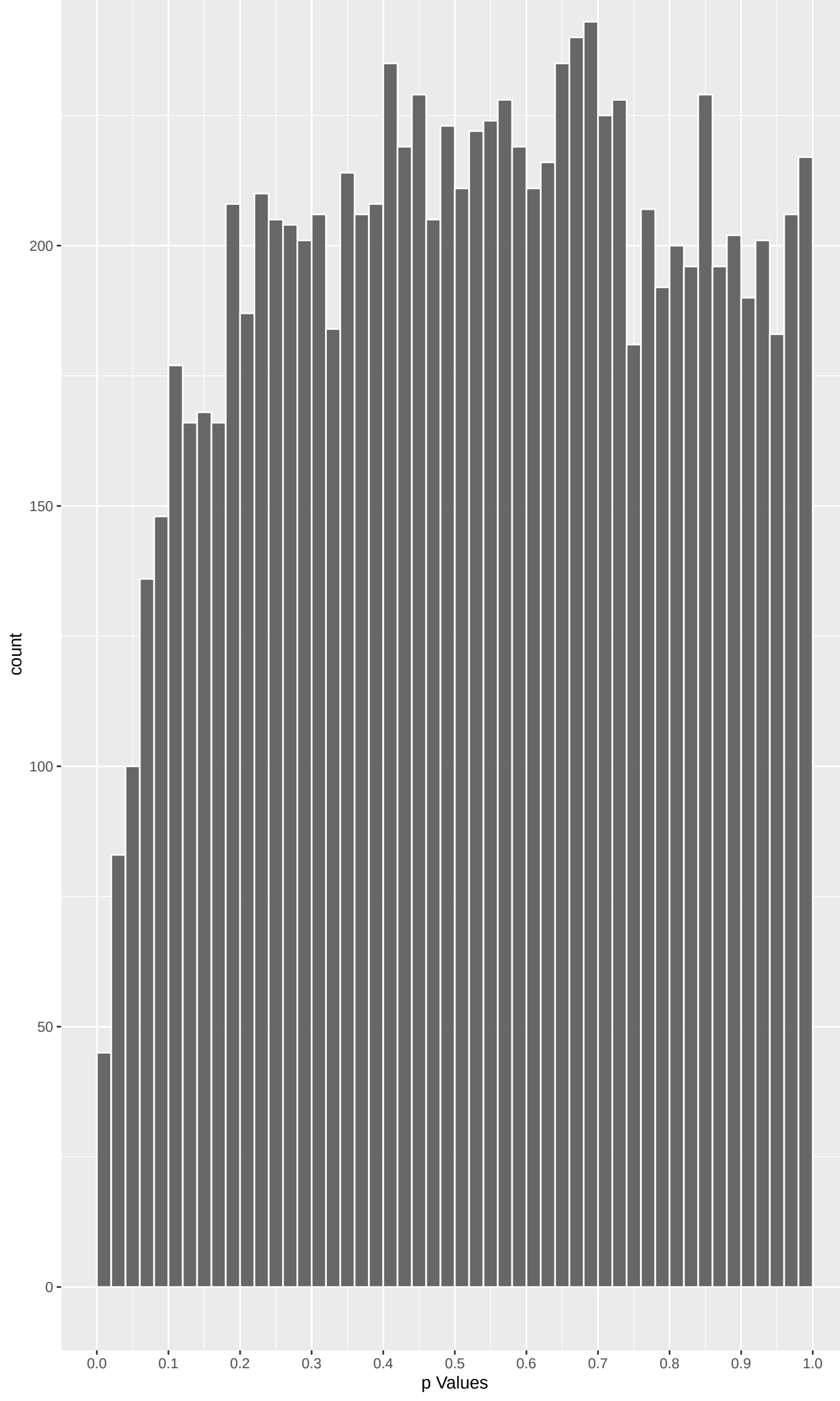
Top genes by Q-Value Permulated

Gene	Rho	N	P	p.adj	permPValue	qValueNoPerm	qValuePerm
A1BG	-0.111644190	123	2.189e-01	9.991e-01	1.986e-01	9.999e-01	1.000e+00
A1CF	-0.024569127	128	7.831e-01	9.991e-01	7.988e-01	9.999e-01	1.000e+00
A2M	-0.160699562	169	3.687e-02	9.991e-01	6.502e-02	9.999e-01	1.000e+00
A2ML1	0.052890311	134	5.439e-01	9.991e-01	4.221e-01	9.999e-01	1.000e+00
A3GALT2	0.013537910	132	8.776e-01	9.991e-01	8.713e-01	9.999e-01	1.000e+00
AGALT	-0.065096551	154	4.225e-01	9.991e-01	4.917e-01	9.999e-01	1.000e+00
AGNT	0.054745291	151	5.043e-01	9.991e-01	4.707e-01	9.999e-01	1.000e+00
AAAS	-0.007388264	132	9.330e-01	9.991e-01	9.663e-01	9.999e-01	1.000e+00
AACS	0.019104271	156	8.129e-01	9.991e-01	7.752e-01	9.999e-01	1.000e+00
AADAC	-0.065142431	107	5.050e-01	9.991e-01	5.055e-01	9.999e-01	1.000e+00
AADACL2	0.019040428	120	8.365e-01	9.991e-01	7.951e-01	9.999e-01	1.000e+00
AADACL3	0.024884968	158	7.563e-01	9.991e-01	7.273e-01	9.999e-01	1.000e+00
AADACL4	-0.023590944	141	7.813e-01	9.991e-01	8.335e-01	9.999e-01	1.000e+00
AADAT	-0.058886959	152	4.711e-01	9.991e-01	5.047e-01	9.999e-01	1.000e+00
AAGAB	-0.010550651	136	9.030e-01	9.991e-01	9.834e-01	9.999e-01	1.000e+00
AAK1	0.140113651	145	9.278e-02	9.991e-01	2.111e-01	9.999e-01	1.000e+00
AAMDC	0.138093817	88	1.995e-01	9.991e-01	3.761e-01	9.999e-01	1.000e+00
AAMP	0.242451740	95	1.792e-02	9.991e-01	9.702e-02	9.999e-01	1.000e+00
AANAT	-0.106992984	138	2.117e-01	9.991e-01	3.568e-01	9.999e-01	1.000e+00
AAR2	0.188222061	133	3.004e-02	9.991e-01	4.217e-02	9.999e-01	1.000e+00
AARD	-0.111535350	105	2.573e-01	9.991e-01	5.427e-01	9.999e-01	1.000e+00
AARS1	-0.064849571	145	4.130e-01	9.991e-01	5.875e-01	9.999e-01	1.000e+00
AARS2	-0.031408831	165	6.888e-01	9.991e-01	6.514e-01	9.999e-01	1.000e+00
AARSD1	0.007615327	145	9.276e-01	9.991e-01	9.879e-01	9.999e-01	1.000e+00
AASDH	0.090375908	165	2.483e-01	9.991e-01	3.356e-01	9.999e-01	1.000e+00

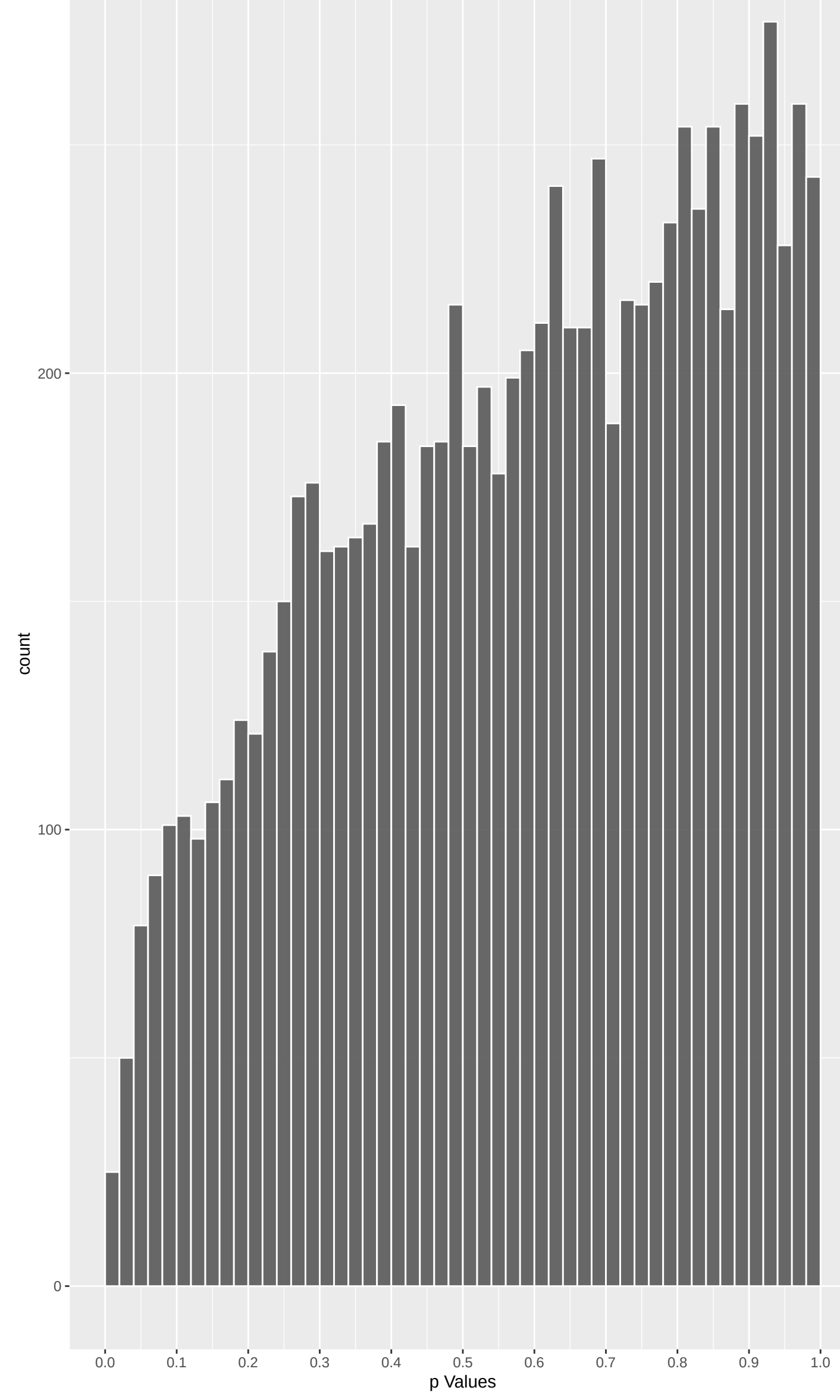
Top genes by Q-Value non-permulated

Gene	Rho	N	P	p.adj	permPValue	qValueNoPerm	qValuePerm
A1BG	-0.111644190	123	2.189e-01	9.991e-01	1.986e-01	9.999e-01	1.000e+00
A1CF	-0.024569127	128	7.831e-01	9.991e-01	7.988e-01	9.999e-01	1.000e+00
A2M	-0.160699562	169	3.687e-02	9.991e-01	6.502e-02	9.999e-01	1.000e+00
A2ML1	0.052890311	134	5.439e-01	9.991e-01	4.221e-01	9.999e-01	1.000e+00
A3GALT2	0.013537910	132	8.776e-01	9.991e-01	8.713e-01	9.999e-01	1.000e+00
AGALT	-0.065096551	154	4.225e-01	9.991e-01	4.917e-01	9.999e-01	1.000e+00
AGNT	0.054745291	151	5.043e-01	9.991e-01	4.707e-01	9.999e-01	1.000e+00
AAAS	-0.007388264	132	9.330e-01	9.991e-01	9.663e-01	9.999e-01	1.000e+00
AACS	0.019104271	156	8.129e-01	9.991e-01	7.752e-01	9.999e-01	1.000e+00
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AADACL2	0.019040428	120	8.365e-01	9.991e-01	7.951e-01	9.999e-01	1.000e+00
AADACL3	0.024884968	158	7.563e-01	9.991e-01	7.273e-01	9.999e-01	1.000e+00
AADACL4	-0.023590944	141	7.813e-01	9.991e-01	8.335e-01	9.999e-01	1.000e+00
AADAT	-0.058886959	152	4.711e-01	9.991e-01	5.047e-01	9.999e-01	1.000e+00
AGAB	-0.010550651	136	9.030e-01	9.991e-01	9.834e-01	9.999e-01	1.000e+00
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AARS2	-0.031408831	165	6.888e-01	9.991e-01	6.514e-01	9.999e-01	1.000e+00
AARSD1	0.007615327	145	9.276e-01	9.991e-01	9.879e-01	9.999e-01	1.000e+00
AASDH	0.090375908	165	2.483e-01	9.991e-01	3.356e-01	9.999e-01	1.000e+00

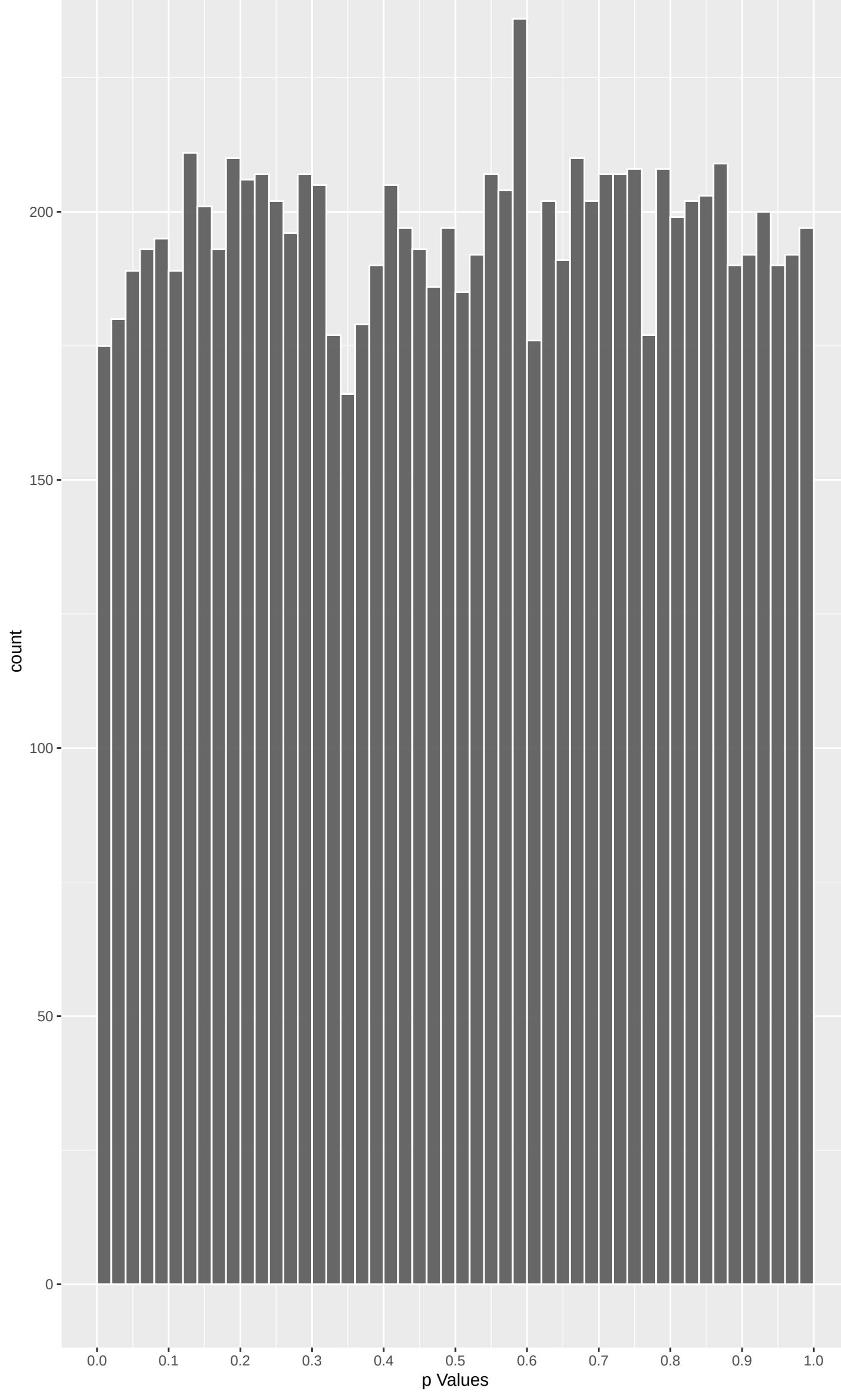
Positive Rho Permulated



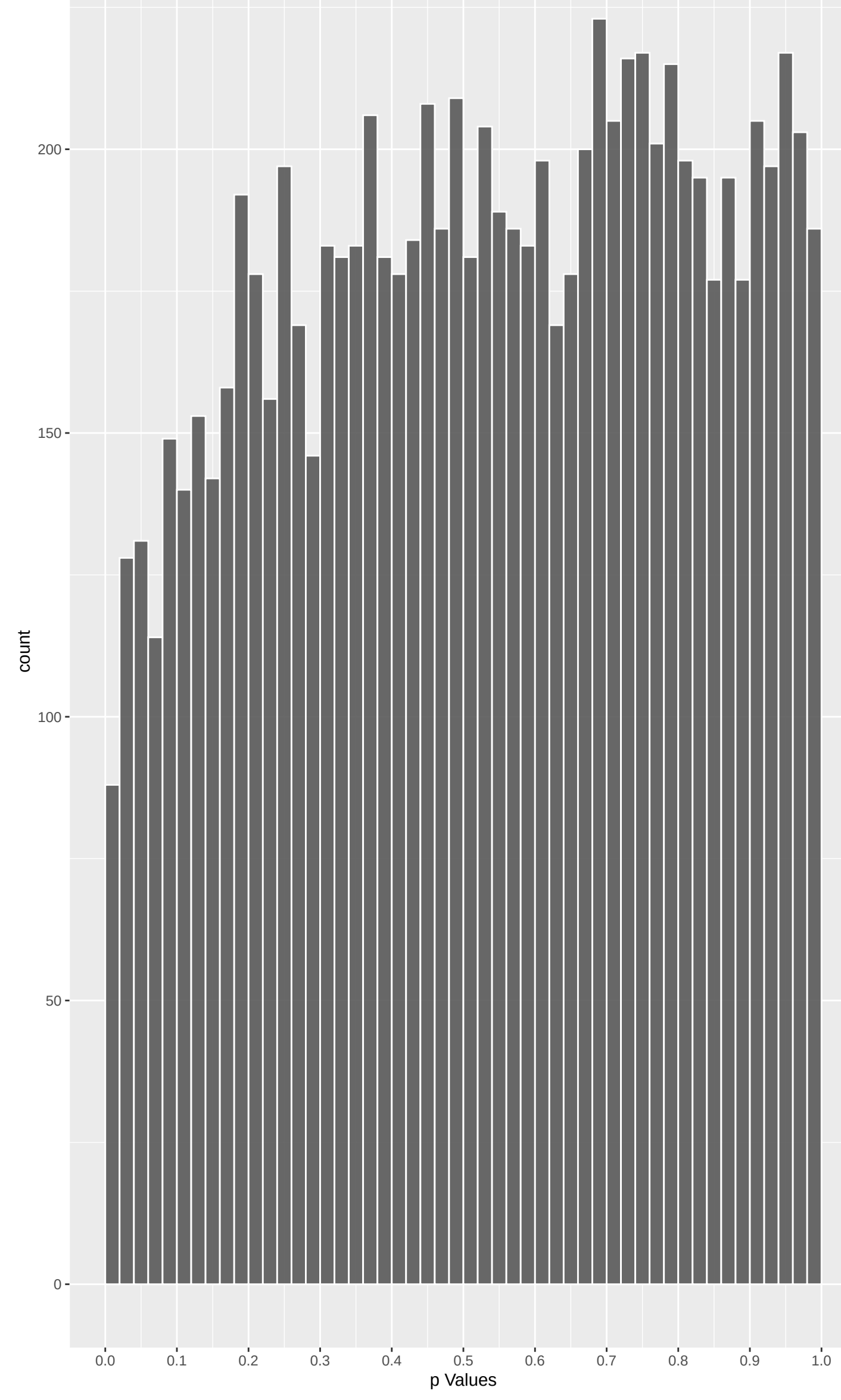
Negative Rho Permulated



Positive Rho Non-permulated



Negative Rho Non-permulated



Top Positive genes by P-value Permulated

	0.2793375	164	2.920e-04	9.991e-01	2.171e-03	9.999e-01	1.000e+00
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ABL2	0.2656947	143	1.340e-03	9.991e-01	7.656e-03	9.999e-01	1.000e+00
CLIP1	0.2715619	165	4.182e-04	9.991e-01	8.686e-03	9.999e-01	1.000e+00
SCFD2	0.2397081	158	2.417e-03	9.991e-01	8.913e-03	9.999e-01	1.000e+00

GO_Biological_Process_2023 Top pathways by permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Sensory Perception Of Smell (GO:0007608)	0.14618253	208	4.158e-13	2.270e-09	OR21P:43 OR5M3:78 OR13G1:89 OR5D3P:102 OR4E2:106 OR10Z1:121.5
Detection Of Chemical Stimulus Involved	0.14697539	129	8.613e-09	2.328e-05	OR21P:43 OR13G1:89 OR4E2:106 OR10Z1:121.5 OR6K3:244.5 OR2T33:304
Detection Of Chemical Stimulus Involved	0.14515771	127	1.691e-08	3.047e-05	OR21P:43 OR13G1:89 OR4E2:106 OR10Z1:121.5 OR6K3:244.5 OR2T33:304
Fatty Acid Metabolic Process (GO:0006631)	-0.14001851	120	1.220e-07	1.649e-04	ACOT8:16 THEM4:38 GNATP:151 LCP1:155 LDLH6B:21 SCPT2:390.5
Monocarboxylic Acid Metabolic Process (G	-0.14347655	96	1.216e-06	1.315e-03	THEM4:38 LDLH6A:64 GNATP:151 LCP1:155 LDLH6B:21 SCPT2:390.5
Peptide Biosynthetic Process (GO:0043043)	0.11155620	153	1.995e-06	1.798e-03	RPS17:34 RPS16:135 AGO2:159.5 RPL9:285 RPL9:302.5 RPL34:367
Cytoplasmic Transport (GO:0022131)	0.14136982	91	3.218e-06	2.468e-03	RPS17:34 RPS16:135 RPL9:283 RPS19:302.5 RPL34:367 RPL23A:411
Brain Development (GO:0007420)	0.10630067	157	4.473e-06	3.023e-03	NDRG4:128 ZIC1:133.5 EML1:31 GRN2C:291 ROB22:293.5 FLNB:383
Macromolecule Biosynthetic Process (GO:0	0.09497648	179	1.222e-05	6.005e-03	RPS17:34 RPS16:135 AGO2:159.5 RPL9:285 RPL9:302.5 GTPBP1:318.5
Sensory Perception Of Chemical Stimulus	0.12823008	98	1.176e-05	6.005e-03	OR5M3:78 OR5D3P:102 CALHM3:140 OR8K1:194 OR5L1:242.5 OR5M9:376.5
Triglyceride Metabolic Process (GO:00066	-0.13099626	43	1.189e-05	6.005e-03	APOC3:29 LCP1:155 LPLN2:206.5 LPLN1:256 GK5:598.5 PRLR:536
Steroid Biosynthetic Process (GO:0006634)	-0.17787624	49	1.665e-05	7.503e-03	ACOT8:16 CYP8B1:163.5 OSBP:266 CYP39A1:311.5 SCPT2:390.5 PRLR:536
Nitrogen Compound Transport (GO:0071705)	-0.09351297	161	4.371e-05	1.818e-02	ABCC2:84.5 CD320:92 SLC13A1:196 SLC22A4:222 SLC38A2:260.5 SLC6A13:372
Gene Expression (GO:0014867)	0.08943954	286	5.663e-05	2.046e-02	RPS17:34 HSC1:1103 HNRNP43:123.5 RPS16:135 AGO2:159.5 ALYREF:162.5
Vitamin Transport (GO:0051180)	-0.13870492	38	5.076e-05	2.046e-02	SLC5A1:34 CD320:92 SLC25A32:128 SLC28B:190 TCM2:285 SLC46A1:598.5
Cellular Lipid Catabolic Process (GO:004	-0.23646837	23	8.667e-05	2.757e-02	ACOT8:16 LPLN2:206.5 LPLN1:256 ACOK1:319 PLBD1:343 PLD1:1397.5
Translation (GO:0006412)	0.07910119	228	8.207e-05	2.757e-02	RPS17:34 RPS16:135 AGO2:159.5 RPL9:285 RPL9:302.5 GTPBP1:318.5
Cell Morphogenesis Involved In Neuron Di	0.12899038	76	1.024e-04	2.915e-02	NEFH:123.5 MAP1B:188.5 DVL1:523 POU4F1:532.5 MNK1:703 APLP2:883.5
Diol Biosynthetic Process (GO:0043312)	-0.28126773	16	9.819e-05	2.915e-02	SPHK2:21 SPTSSB:40.5 ACER1:321 SPTLC1:496.5 DHRF:542.5 SPHK1:900
Neuron Projection Morphogenesis (GO:0048	0.09432961	140	1.196e-04	3.233e-02	SRGAP3:82 NEFH:123.5 MAP1B:188.5 EPHA4:251.5 SGK1:261.5 SHANK3:429
Fatty Acid Catabolic Process (GO:0009062	-0.14210875	60	1.409e-04	3.493e-02	ACOT8:16 HSC1:1103 HNRNP43:123.5 RPS16:135 AGO2:159.5 LPLN1:256 ACOK1:319
Serine Family Amino Acid Metabolic Proce	-0.30475009	13	1.421e-04	3.493e-02	GLC1:320 AOC1:831 SPRR1:210 CTH1:172 GYAT:199.5 BART:201.5
Lipid Biosynthetic Process (GO:0008610)	-0.13224147	68	1.643e-04	3.554e-02	LCP1:155 SCPT2:390.5 MVR:534.5 PRLR:536 ACACB:634 CYP17A1:625
Sphingoid Biosynthetic Process (GO:00465	-0.31482171	12	1.593e-04	3.554e-02	SPHK2:21 SPTSSB:40.5 ACER1:321 SPTLC1:496.5 SPHK1:900 ASAH2:908
Sphingosine Biosynthetic Process (GO:004	-0.31482171	12	1.593e-04	3.554e-02	SPHK2:21 SPTSSB:40.5 ACER1:321 SPTLC1:496.5 SPHK1:900 ASAH2:908
NADH Dehydrogenase Complex Assembly (GO	-0.15405705	49	1.922e-04	3.849e-02	EC5IT:70.5 NDUFB10:271.5 NDUFAF1:469 NUBPL:474.5 NDUFA3:606 NDUFC2:614.5
Mitochondrial Respiratory Chain Complex	-0.15405705	49	1.922e-04	3.849e-02	EC5IT:70.5 NDUFB10:271.5 NDUFAF1:469 NUBPL:474.5 NDUFA3:606 NDUFC2:614.5
Positive Regulation Of Cell Migration (G	-0.06587634	267	2.214e-04	4.127e-02	PDGFRB:27 EPHA2:60 MYADM:65 CTSH:90.5 SEMA7A:148 GRB7:295.5
Positive Regulation Of Leukocyte Activat	-0.26889495	20	2.173e-04	4.127e-02	PDGFRB:27 EPHA2:60 MYADM:65 CTSH:90.5 SEMA7A:148 GRB7:295.5
Embryonic Limb Morphogenesis (GO:003528	0.13232552	34	2.342e-04	4.230e-02	NTU1:217 WDRP3:825 CREBBP:817 TRAP2A:936 TBX2:113.5 ZNF358:1164.5
Mitochondrial Respiratory Chain Complex	-0.11687195	82	2.568e-04	4.340e-02	EC5IT:70.5 NDUFB10:271.5 NDUFAF1:469 NUBPL:474.5 CCHD:70.6 NDUFA3:606
protein-RNA Complex Assembly (GO:0022618	0.08949784	741	2.494e-04	4.340e-02	AGO1:32 TARBP2:423 AGO2:159.5 SART1:195 RPS19:302.5 LUC7L2:354
Wnt Signaling Pathway (GO:0016055)	0.12170534	14	2.979e-04	4.350e-02	FZD8:59 CTGF:11.223 CDC08C:223 WNT7B:347 FZD3:487 DVL1:523
acyl-CoA Metabolic Process (GO:0006637)	-0.14066240	28	3.800e-04	4.350e-02	ACOT8:16 MCEE:445 ACACB:624 OGDH:631 SUCCLG:794 SUCLA2:1257
Amide Biosynthetic Process (GO:0043604)	-0.14905511	50	2.669e-04	4.350e-02	SPTSSB:40.5 MCEE:445 ACACB:624 OGDH:631 SUCCLG:794 SUCLA2:1257
Amide Transport (GO:0042388)	-0.18457077	31	3.678e-04	4.350e-02	SLC5A6:34 SLC25A32:128 OSBP:266 GIPC1:313 DISP1:438 TAP1:576.5
Cholesterol Transport (GO:0030301)	-0.14404802	51	3.769e-04	4.350e-02	APOC3:29 OSBP:266 STARD5:349 SLC41:403 CDM2:728.5 SLC16:187.5
Dicarboxylic Acid Metabolic Process (GO:	-0.14840342	49	3.279e-04	4.350e-02	ACOT8:16 SLC25A32:128 GLC1:320 LZHQDH:442.5 DHRF:542.5 SLC46A1:598.5
Import Into Cell (GO:0009867)	-0.17046163	37	3.345e-04	4.350e-02	SLC5A6:34 SLC22A4:222 SLC16A2:242 SLC6A1:372 SLC6A4:598.5 SLC6A6:738
mRNA Processing (GO:0006397)	0.07239027	205	3.650e-04	4.350e-02	HNRNP43:123.5 ALYREF:162.5 SART1:195 RBMX:238.5 SNRPB:1242.5 DDX15:283

EnrichmentHsSymbolsFile2 Top pathways by permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_OLFACTORY_SIGNALING_PATHWAY	0.16513987	390	5.512e-29	3.509e-25	ORS2B2:19.5 OR52W1:29 OR2AT4:62 OR5A1:78 OR13G1:89 OR11G2:104
KEGG_OLFACTORY_TRANSDUCTION	0.15889558	380	2.536e-26	8.234e-23	ORS2B2:19.5 OR52W1:29 CAMK2B:42 OR2AT4:62 OR5M3:78 OR13G1:89
REACTOME_SENSORY_PERCEPTION	0.12097937	605	3.978e-24	8.610e-21	ORS2B2:19.5 OR52W1:29 OR2AT4:62 MYH9:69 OR5M3:78 OR13G1:89
REACTOME_DEVELOPMENTAL_BIOLOGY	0.08326744	1143	3.903e-21	6.337e-18	KRTAP10:13 SPRR2B:7 AB128 RPS17:34 MAP2K2:50 MYH6:69
HOUPKE_HOUSEKEEPING_GENES	0.06709069	1100	8.182e-14	1.063e-10	CBX1:19.5 DDXT7:35 PDCDC36:5 TSS3:39 PURB:41 MAP2K2:50
REACTOME_SIGNALING_BY_WNT	0.11693456	322	5.894e-13	6.379e-10	AGO1:32 FZD8:59 HSC1:103 HSC13:127 H2BCA1:411 AGO2:159.5
HSIAO_LIVER_SPECIFIC_GENES	-0.12999976	240	4.388e-12	4.071e-09	APOC3:29 ABCC2:84.5 NR1H4:130.5 ALDH1A1:139.5 LPLN2:206.5 AZGP1:369
CARRILLOREXACH_HEPATOBLASTOMA_VS_NORMAL	-0.06349143	1046	5.424e-12	4.403e-09	THEM4:38 APOC3:29 ANKOS:25 SLC6A16:53 ETNPPL:99 EPHA2:60
REACTOME_INFECTION_DISEASE	0.06151647	933	2.342e-10	1.690e-07	MBX1:19.5 RPS17:34 MAP2K2:50 MYH9:69 TUBA1A:71.5 IPNA14:74
REACTOME_ACTIVATION_OF_ANTERIOR_HOX_GENE	0.16697291	116	5.322e-10	3.456e-07	HSC1:1103 HSC13:127 H2BCA1:411 H2AC1A:261.5 H2AC1B:308.5 ASH2L:394
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.07574312	570	7.112e-10	4.199e-07	ABL2:8 RPS17:34 MAP2K2:50 MYH9:69 TUBA1A:71.5 SRGAP3:82
REACTOME_METABOLISM_OF_LIPIDS	-0.06600441	726	9.558e-10	5.172e-07	ACOT8:16 SPHK2:21 STARD5:34 THEM4:38 SPTSSB:40.5 MTMR6:44.5
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	0.06418692	788	1.044e-09	5.214e-07	AGO1:32 RPS17:34 CAMK2B:42 TUBA1A:71.5 CALR:90 TXN2:95.5
REACTOME_RHO_GTPASES_ACTIVATE_PKNS	0.18409103	89	1.961e-09	9.096e-07	MYH9:69 YWHAQ:94 HSC1:103 HSC13:127 H2BCA1:411 H2AC1A:261.5
REACTOME_REGULATION_OF_EXPRESSION_OF_SLI	0.13242352	169	2.937e-09	1.271e-06	RPS17:34 RPS16:135 PSMA4:225 PSMA4:225 RPL9:283 ROB2:293.5
HOSHIDA_LIVER_CANCER_SUPPRESSION_SS	-0.04141267	260	7.793e-09	3.163e-06	GGH50 ALDH1A1:139.5 LPLN2:206.5 SLC6A2:242 LPLN1:256 ADK:774
CAIRO_LIVER_DEVELOPMENT_DN	-0.11329995	217	9.121e-09	3.494e-06	CTSH:90.5 CAMP:104 LCP1:155 ANK1:278 LPLN2:206.5 SLC16A2:242
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	0.13116255	214	1.198e-08	4.322e-06	ABL2:8 RPS17:34 SRGAP3:82 RPS16:135 KRTAP10:13 PSMA7:258
REACTOME_KERATINIZATION	0.12165690	214	1.391e-08	4.753e-06	KRTAP10:13 SPRR2B:7 KRTAP10:13 KRT81:91.5 KRTAP10:13 KRTAP5:8.130.5
REACTOME_BETA_CATENIN_INDEPENDENT_WNT_SI	0.13609902	145	1.568e-08	5.092e-06	AGO1:32 FZD8:59 AGO2:159.5 PDE6G:165.5 TNRC6B:176.5 CAMK2A:192
REACTOME_METABOLISM_OF_STEROIDS	-0.13195589	152	2.006e-08	6.203e-06	ACOT8:16 STARD6:31 TM7SF2:116 NR1H4:130.5 CYP8B1:163.5 KRB1A1:1213
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	0.10493804	233	3.539e-08	1.045e-05	AGO1:32 HSC1:103 HSC13:127 H2BCA1:411 AGO2:159.5 CBX2:173
REACTOME_INFECTION	0.12623219	179	4.448e-08	1.256e-05	MYH9:69 ACTR1:190 ARPC4:276 ARPC3:417.5 VAV1:478 FGR:488.5
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	-0.11968025	145	2.734e-08	1.553e-05	ETNPPL:99 CYP8B1:163.5 MYRIP:25.5 SLC16A2:242 AZGP1:369 SCPT2:390.5
CAIRO_HEPATOBLASTOMA_DN	-0.09717051	262	6.362e-08	1.653e-05	ETNPPL:99 EPHA2:60 MGLL:98 LEPR:182.5 LPLN2:206.5 VPR1:246
REACTOME_RHO_GTPASE_EFFECTORS	0.06729911	312	1.192e-07	2.848e-05	CLU1P:9 CIT16 MYH9:69 TUBA1A:71.5 YWHAQ:94 HSC1:1103
REACTOME_OXIDATIVE_STRESS_INDUCED_SENESC	0.14145624	117	1.272e-07	2.848e-05	AGO1:32 HSC1:103 HSC13:127 H2BCA1:411 AGO2:159.5 TNRC6B:176.5
REACTOME_GLYCOXYLATE_METABOLISM_AND_GLYCI	-0.27889902	30	1.242e-07	2.848e-05	DBT:201 DLD:242 GLDC5:63.5 UAS:566 OGDH:631 PHOAZ:637.5
REACTOME_ESR_MEDIATED_SIGNALING	0.10506695	214	1.209e-07	2.848e-05	AGO1:32 HSC1:103 HSC13:127 H2BCA1:411 AGO2:159.5 TNRC6B:176.5
REACTOME_TRANSCRIPTIONAL_REGULOMEN_INVOLVED	0.15672222	92	2.061e-07	4.460e-05	AGO1:32 HSC1:103 HSC13:127 H2BCA1:411 TNRC6B:176.5 SETD1B:246.5
REACTOME_ESTROGEN_DEPENDENT_GENE_EXPRESS	0.12340408	145	2.951e-07	5.961e-05	AGO1:32 HSC1:103 HSC13:127 H2BCA1:411 AGO2:159.5 TNRC6B:176.5
WP_CYTOSOLASMIC_RIBOSOMAL_PROTEINS	0.16018754	86	2.845e-07	5.861e-05	RPS17:34 RPS16:135 RPL9:283 RPS19:302.5 RPL34:367 RPL23A:411
HSIAO_HOUSEKEEPING_GENES	0.07661425	381	2.978e-07	5.861e-05	RPS17:34 MYH9:69 HNT1:86 YWHAQ:94 RPS16:135 SLC25A6:143
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	-0.22288267	44	3.137e-07	5.992e-05	DBT:201 DLD:242 ACAD8:286 HIBCH:324 MCEE:445 BCAT2:474.5
REACTOME_SRP_DEPENDENT_COTRANSLATIONAL_P	0.13893881	111	4.306e-07	7.990e-05	RPS17:34 RPS16:135 RNP2:213.5 RPL9:283 RPS19:302.5 RPL34:367
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	0.13040311	125	4.837e-07	8.725e-05	HSC1:103 HSC13:127 H2BCA1:411 PSMA4:225 PSMA7:258 H2AC1A:261.5
MARTENS_TRETINOLIN_RESPONSE_DN	0.05394735	751	5.370e-07	9.425e-05	CLU1P:9 ARL6P4:33 PURB:41 ABT1:67 HNT1:86 CALR:90
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	-0.09289152	245	6.010e-07	1.002e-04	SLC22A8:8 SLC6A12:12 SLC5A6:34 SLC6A1:372 SLC13A1:196 SLC22A6:204
KIM_BIPOLAR_DISORDER_OLCODENDROCYTIC_DEN	0.06578659	655	5.326e-07	1.002e-04	RPS17:34 CAMK2B:42 PDCDC36:5 HNT1:86 YWHAQ:94 NDRG4:129
REACTOME_TCF_DEPENDENT_SIGNALING_IN_RESP	0.09592966	226	6.908e-07	1.094e-04	FZD8:59 HSC1:103 HSC13:127 H2BCA1:411 CDC08C:223 TCF7L1:1223

DisGeNET Top pathways by permulation

Vomiting	-0.10445384	175	1.964e-06	2.100e-03	DBT201.DLD242.SLC1A5:277.5.CPM3:385.TON2:988.PPK4:566.5
Lethargy	-0.13410090	104	2.359e-06	2.106e-03	DBT201.DLD242.SLC25A20:355.TON2:988.DPYD:461.5.NDUFA1:469
Diabetes Mellitus (Insulin-Dependent)	-0.04379366	1033	2.653e-06	2.171e-03	PDE1A:26.APOC3:29.EPHA2:60.IL37:66.TM6SF2:68.5.CAMP:104
Hyperlipidemia	-0.09241970	210	4.124e-06	3.116e-03	MFAP5:1.NR1H4:130.5.BCAS3:62.5.CPM1:134.LEPR:182.5.AKR1B1:213.OLR1:223.5
Coronary Artery Disease	-0.04513434	898	5.749e-06	4.028e-03	MFAP5:1.APOC3:29.BCAS3:62.5.HIBCH:324.MCEE:445.MYO9B:84.5.ELL:95.5
Metabolic Diseases	-0.06684489	392	6.152e-06	4.028e-03	APOC3:29.ACAD8:286.IL37:66.TM7SF2:116.NR1H4:130.5.LEPR:182.5
Nonalcoholic Steatohepatitis	-0.09396535	192	7.473e-06	4.987e-03	TM6SF2:68.5.LAMA1:97.NR1H4:130.5.CYP8B1:163.5.LEPR:182.5.OLR1:223.5
Hypercholesterolemia, Familial	-0.10651306	145	8.959e-06	5.739e-03	APOC3:29.ACAD8:286.OLR1:223.5.ITGB3:388.5.MTP:407.5.ABCA1:514
Hypertriglyceridemia	-0.08071665	141	8.715e-06	5.739e-03	APOC3:29.TM6SF2:68.5.NR1H4:130.5.LPN1:256.MTP:407.5.ABCA1:514
Liver Cirrhosis	-0.05066524	625	1.5900e-05	8.216e-03	SLC22A8:8.TM6SF2:68.5.HES6:107.TM7SF2:116.NR1H4:130.5.A2M1:178
Hepatitis, Toxic	-0.08725355	198	2.407e-05	1.028e-02	SLC22A8:8.GGH50.ABCG2:84.5.NR1H4:130.5.ECHDC1:159.5.CYP8B1:163.5
Chemical and Drug Induced Liver Injury	-0.08725355	198	2.407e-05	1.028e-02	SLC22A8:8.GGH50.ABCG2:84.5.NR1H4:130.5.ECHDC1:159.5.CYP8B1:163.5
Chemically Induced Liver Toxicity	-0.08725355	198	2.407e-05	1.028e-02	SLC22A8:8.GGH50.ABCG2:84.5.NR1H4:130.5.ECHDC1:159.5.CYP8B1:163.5
Drug-Induced Acute Liver Injury	-0.08725355	198	2.407e-05	1.028e-02	SLC22A8:8.GGH50.ABCG2:84.5.NR1H4:130.5.ECHDC1:159.5.CYP8B1:163.5
Cerebral Hemorrhage	-0.15194022	64	2.655e-05	1.043e-02	KRT11:3.DH4:22.ITGA1V:130.5.COL4A2:24.ITGB3:338.5.BAX:411.5
Coronary Arteriosclerosis	-0.04681698	706	2.655e-05	1.043e-02	APOC3:29.PRKY:114.NR1H4:130.5.COL4A2:24.AKR1B1:213.OLR1:223.5
Liver diseases	-0.05146044	555	3.837e-05	1.494e-02	APOC3:29.TM6SF2:68.5.ABCG2:84.5.TM7SF2:116.NR1H4:130.5.ALDH1A1:139.5
Hepatitis, Drug-Induced	-0.08386215	202	4.142e-05	1.507e-02	SLC22A8:8.GGH50.ABCG2:84.5.NR1H4:130.5.ECHDC1:159.5.CYP8B1:163.5
Arteriosclerosis	-0.03893312	988	4.317e-05	1.514e-02	APOC3:29.EPHA2:60.IL37:66.TM6SF2:68.5.CAMP:104.TM7SF2:116
Mitochondrial Diseases	-0.06206834	303	5.234e-05	1.773e-02	SDHC:42.ECSIT:70.5.NAXE:173.CERS6:187.DLD242.NDUFB1:201.5
Pre-Eclampsia	-0.0702182	148	5.695e-05	1.854e-02	GOM1:3.TPBG:67.LEPR:182.5.OLR1:223.5.APLNR:394.5.ADM:39
Arrhythmia	-0.04827740	575	8.547e-05	2.708e-02	NR1H4:130.5.HK1:101.TM7SF2:116.SLC22A2:782.MR3:29
Diabetes Mellitus (Insulin-Dependent)	-0.04664294	144	9.469e-05	2.708e-02	ADAM28:22.APOC3:29.EPHA2:60.IL37:66
Adverse reaction to drug	-0.12570114	80	1.029e-04	2.854e-02	ABCC2:84.5.CC:29.5.BZM:31.5.TON2:988.DHFR:54.5.MFE2L2:73.5
Drug toxicity	-0.12570114	80	1.029e-04	2.854e-02	ABCC2:84.5.CC:29.5.BZM:31.5.TON2:988.DHFR:54.5.MFE2L2:73.5
Periodontal Diseases	-0.09998283	128	9.589e-05	2.854e-02	CAMP:104.AMX1:39.TN2.DEBF1:348.CTSD:417.BDNF:521.5.C1S:815
Vascular Diseases	-0.0852591	377	1.027e-04	2.854e-02	APOC3:29.APM1:95.5.TM7SF2:116.AKTIR1:153.AKR1B1:213.OLR1:223.5
Anorexia	-0.11389978	95	1.267e-04	3.191e-02	LEPR:182.5.IAPP:30.5.MAGEL2:40.5.HTR3:50.5.BDNF:521.5.SLC6A1:598.5
Hyperinsulinemia	-0.0918294	361	1.203e-04	3.191e-02	APOC3:29.CTSH:90.5.ODF1:113.CR1B:177.LEPR:182.5.LPN1:256
Myocardial Infarction	-0.08317725	899	1.239e-04	3.191e-02	APOC3:29.BCAS3:62.5.TM6SF2:68.5.IF3:79.TASR2:80.5.TM7SF2:116
Comatose	-0.12808436	71	1.921e-04	4.716e-02	NAXE:173.DBT:201.SLC25A20:355.NDUFA1:469.NUPL1:474.5.TM126:667.5